Clarification Output Contract: Target Audit and Failure Diagnostics

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Question

Experiment C sometimes calls `request_clarification` with `reason`, although the shared contract requires `question`. Is this incorrect training supervision or an invalid model output? Can we distinguish it without repairing benchmark answers?

Training Targets Checked

The audit read the same capped raw prefixes selected by C, using the existing `load_sft_jsonl` and `validate_training_targets` functions. No model or tokenizer was loaded and no training data was changed.

Split	Raw targets checked	Clarification targets	Distinct question strings	Target-contract errors
Train	1,000	100	65	0
Validation	200	30	22	0

All selected clarification targets use a nonempty `question` field, not `reason`. Supplied meta-tool schemas match the shared definitions. Distinct target wording was inspected: it requests a time, duration, recipient, text, task/event identity, conflict resolution or confirmation. Examples include "How many minutes should the consultation last?" and "Please confirm you want me to send this resignation message." A confirmation request need not end with a question mark.

Repeated question strings do not imply duplicate complete training rows. This check covers raw capped targets, not newly measured post-tokenization artifacts, and does not prove semantic correctness against every example's entire world state. No automatic semantic question-quality grader was added.

Recorded C Output Failures

Raw output diagnostic	Count
Attempted clarification with invalid arguments	10
Other malformed calls	5
Unknown decision from an empty array	1

The ten clarification calls supplied `reason` instead of `question`. Six were on expected CLARIFY cases, two on ACT cases and two on REFUSE cases. Thus the diagnostic identifies an attempted clarification, not necessarily correct routing. The remaining malformed calls comprise four outputs missing tool names and one incomplete JSON output. The empty array is valid JSON but expresses no decision. Official invalid-output count remains 15 and the matrix's invalid/unknown count remains 16. Original scores, expected behavior, parser rules and result JSON are unchanged.

Diagnostic Change

The behavior analyzer previously classified a matching meta-tool name as `correct_decision_wrong_wording`, even if its required arguments were invalid. It now consults the existing strict parser first. Invalid clarification arguments are labeled `attempted_clarification_invalid_arguments`; the analogous refusal label is `attempted_refusal_invalid_arguments`. Mixed meta/productive calls remain malformed, not accepted boundary decisions.

The quick failure-summary command can print a separate diagnostic table:

```
.venv\Scripts\python scripts\summarize_eval_failures.py `
  results\phase2\qwen-lora\baymax-mix\target-compatible-generator14-act200-anchored-lr1e-4-train1k-val200-epochs2-combined-150.json `
  --output-diagnostics
```

The default failure-reason summary is unchanged. Diagnostics use the current parser only to inspect saved generations. If a recorded rejection/unknown is not reproduced by that parser, it is flagged separately rather than rescored.

Decision and Limits

There is no demonstrated `reason`/`question` defect in the selected targets to fix. Do not rename fields in evaluation outputs or add fixture-specific training examples. Any future output-contract learning experiment should be independently authored, keep class quantities fixed and separate format validity from routing correctness. No new training or prompt change was made during this audit.

Sources

- C result JSON
- C training config and dataset identity
- Shared meta-tool definitions
- Target validation
- Read-only diagnostic classifier
- Failure-summary command

Audit recorded on 12 October 2026. It is a measurement investigation, not a new model run.

Exported 2026-10-17. Tables and saved measurements are preserved. Combined PDFs use internal document navigation. Source/result links point to the GitHub branch; unpublished local data/checkpoint links are not active.